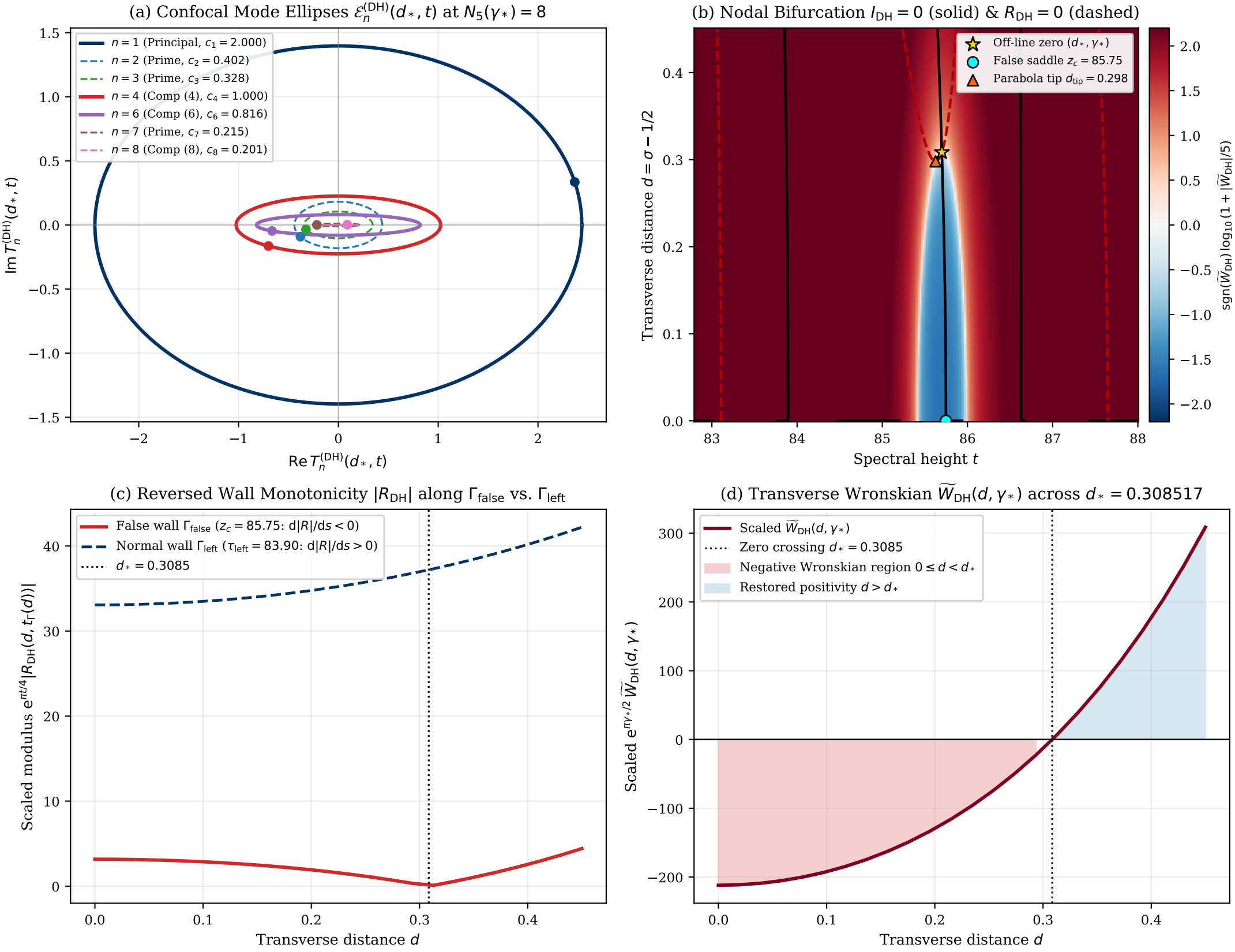

(a) Confocal Mode Ellipses $\mathcal{E}_n^{(\mathrm{DH})}(d_*, t)$ at $N_5(\gamma_*) = 8$
$n = 1$ (Principal, $c_1 = 2.000$)
$n = 2$ (Prime, $c_2 = 0.402$)
$n = 3$ (Prime, $c_3 = 0.328$)
$n = 4$ (Comp (4), $c_4 = 1.000$)
$n = 6$ (Comp (6), $c_6 = 0.816$)
$n = 7$ (Prime, $c_7 = 0.215$)
$n = 8$ (Comp (8), $c_8 = 0.201$)
$\mathrm{Im}\, T_n^{(\mathrm{DH})}(d_*, t)$
$\mathrm{Re}\, T_n^{(\mathrm{DH})}(d_*, t)$
(b) Nodal Bifurcation $I_{\mathrm{DH}} = 0$ (solid) & $R_{\mathrm{DH}} = 0$ (dashed)
Off-line zero (d_*, γ_*)
False saddle $z_c = 85.75$
Parabola tip $d_{\mathrm{tip}} = 0.298$
Transverse distance $d = \sigma - 1/2$
Spectral height t
$\mathrm{sgn}(\widetilde{W}_{\mathrm{DH}}) \log_{10}(1 + |\widetilde{W}_{\mathrm{DH}}|/5)$
(c) Reversed Wall Monotonicity $|R_{\mathrm{DH}}|$ along Γ_{false} vs. Γ_{left}
False wall Γ_{false} ($z_c = 85.75$: $\mathrm{d}|R|/\mathrm{d}s < 0$)
Normal wall Γ_{left} ($\tau_{\mathrm{left}} = 83.90$: $\mathrm{d}|R|/\mathrm{d}s > 0$)
$d_* = 0.3085$
Scaled modulus $e^{\pi t/4}|R_{\mathrm{DH}}(d, t_\Gamma(d))|$
Transverse distance d
(d) Transverse Wronskian $\widetilde{W}_{\mathrm{DH}}(d, \gamma_*)$ across $d_* = 0.308517$
Scaled $\widetilde{W}_{\mathrm{DH}}(d, \gamma_*)$
Zero crossing $d_* = 0.3085$
Negative Wronskian region $0 \le d < d_*$
Restored positivity $d > d_*$
Scaled $e^{\pi\gamma_*/2}\widetilde{W}_{\mathrm{DH}}(d, \gamma_*)$
Transverse distance d